Review I

1. Section 6.1
   a. Area between a curve y = f(x) and the x-axis
   b. Area between two curves y = f(x) and y = g(x)
      i. Approximation by *vertical* strips
      ii. $Area = \int_a^b [\, f(x) - g(x) \,]\, dx$
   c. Area between two curves x = G(y) and x = F(y)
      iii. Approximation by *horizontal* strips
      iv. $Area = \int_c^d [\, G(y) - F(y) \,]\, dy$
   d. Consumer Surplus
   e. Representative Problems: 1, 2, 7, 8, 13, 14, 27, 28

2. Section 6.2
   a. Volume of solid with cross-sectional area A(x)
      i. $Volume = \int_a^b A(x)\, dx$
   b. Volume of solid of revolution obtained by revolving region R about x-axis
      i. Region R bounded between the curve y = f(x) and x-axis
         a) *disk method*
         b) $Volume = \pi \int_a^b [\, f(x) \,]^2\, dx$
      ii. Region R bounded between the curve y = f(x) and y = g(x)
         a) *washer method*
         b) $Volume = \pi \int_a^b (\, [\, f(x) \,]^2 - [\, g(x) \,]^2 \,)\, dx$
   c. Volume of solid of revolution obtained by revolving region R about y-axis
      i. Region R bounded between the curve x = G(y) and y-axis
         a) *disk method*
         b) $Volume = \pi \int_a^d [\, G(y) \,]^2\, dy$
      ii. Region R bounded between the curve x = G(y) and x = F(y)
         a) *washer method*
         b) $Volume = \pi \int_c^d (\, [\, G(y) \,]^2 - [\, F(y) \,]^2 \,)\, dy$
   d. Extensions to revolving region R about other axes
   e. Representative Problems: 2, 14, 15, 19, 31, 32, 41, 42, 47, 49

3. Section 6.3
   a. Shell method for volume of solid of revolution obtained by revolving region R about y-axis
      i. Region R bounded between the curve y = f(x) and x-axis
         a) $Volume = \int_a^b 2\pi x\ f(x)\, dx$
      ii. Region R bounded between the curve y = f(x) and y = g(x)
         a) $Volume = \int_a^b 2\pi x\, [\, f(x) - g(x) \,]\, dx$
   b. Shell method for volume of solid of revolution obtained by revolving region R about x-axis
      i. Region R bounded between the curve x = G(y) and y-axis
         a) $Volume = \int_c^d 2\pi y\ G(y)\, dy$
      ii. Region R bounded between the curve x = G(y) and x = F(y)
         a) $Volume = \int_c^d 2\pi y\, [\, G(y) - F(y) \,]\, dy$
   c. Extensions to revolving region R about other axes
   d. Representative Problems: 11, 12, 19, 20, 26, 28, 30

4. Section 6.4
   a. Arc length of a curve C given by y = f(x)
      i. $s = \int_a^b \sqrt{1 + [\, f'(x) \,]^2}\ dx$
   b. Arc length of a curve C given by x =g(y)
      i. $s = \int_c^d \sqrt{1 + [\, g'(y) \,]^2}\ dy$

c. Surface area of a surface of revolution obtained by revolving a curve C about an axis
   i. Curve C given by y = f(x) revolving about x-axis
      a) $S = \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2}\, dx$
   ii. Curve C given by y = f(x) revolving about y-axis
      a) $S = \int_a^b 2\pi x \sqrt{1 + [f'(x)]^2}\, dx$

d. Surface area of a surface of revolution obtained by revolving a curve C about an axis
   i. Curve C given by x = g(y) revolving about y-axis
      a) $S = \int_c^d 2\pi g(y) \sqrt{1 + [g'(y)]^2}\, dy$
   ii. Curve C given by x = g(y) revolving about x-axis
      a) $S = \int_c^d 2\pi y \sqrt{1 + [g'(y)]^2}\, dy$

e. Extensions to revolving C about other axes

f. Representative Problems: 4, 6, 19, 22

5. Section 6.5
   a. Work done by constant force (physics definition)
   b. Work done by variable force y = F(x) over interval [a,b]
      i. $Work = \int_a^b F(x)\, dx$
      ii. Spring problems (where Hooke's Law applies)
      iii. Lifting leaking container problems
      iv. Lifting rope problems (where weight of rope per unit is significant)
      v. Pumping out liquid containers
   c. Fluid Force
      i. Force on plane at constant depth: F = pressure · area = ρ · h · area
      ii. Force on plane at variable depth with cross-sectional width L(h)
         a) $F = \int_a^b \rho\, h\, L(h)\, dh$
   d. Mass and moments of thin homogenous plate (lamina) R
      i. Plate R bounded between two curve y = f(x) and y = g(x)
         a) Mass = $M = \int_a^b dm = \int_a^b [\, f(x) - g(x)\,]\, dx$
         b) Moment about y-axis = $M_y = \int_a^b \tilde{x}\, dm = \int_a^b x\,[\, f(x) - g(x)\,]\, dx$
         c) Moment about x-axis = $M_x = \int_a^b \tilde{y}\, dm = \int_a^b \frac{(\, f(x) + g(x)\,)}{2}\,[\, f(x) - g(x)\,]\, dx$
   e. Coordinates of centroid
      i. $\bar{x} = \frac{M_y}{M}\ ,\ \bar{y} = \frac{M_x}{M}$
   f. Theorem of Pappus
   g. Representative Problems: 6, 7, 10, 11, 14, 17, 19, 21, 23, 25, 34, 38, 39

6. Section 7.1
   a. Basic anti-differentiation formulas
      i. Power rule: case n ≠ -1 and case n=-1 (logarithm case)
      ii. Exponential and trigonometric functions
      iii. Inverse trigonometric functions
   b. Substitution (reversing the effects of the chain rule)
   c. Using Tables of Integrals
   d. Representative Problems: 1, 2, 6, 7, 15, 19, 27, 28, 34, 42, 46, 47